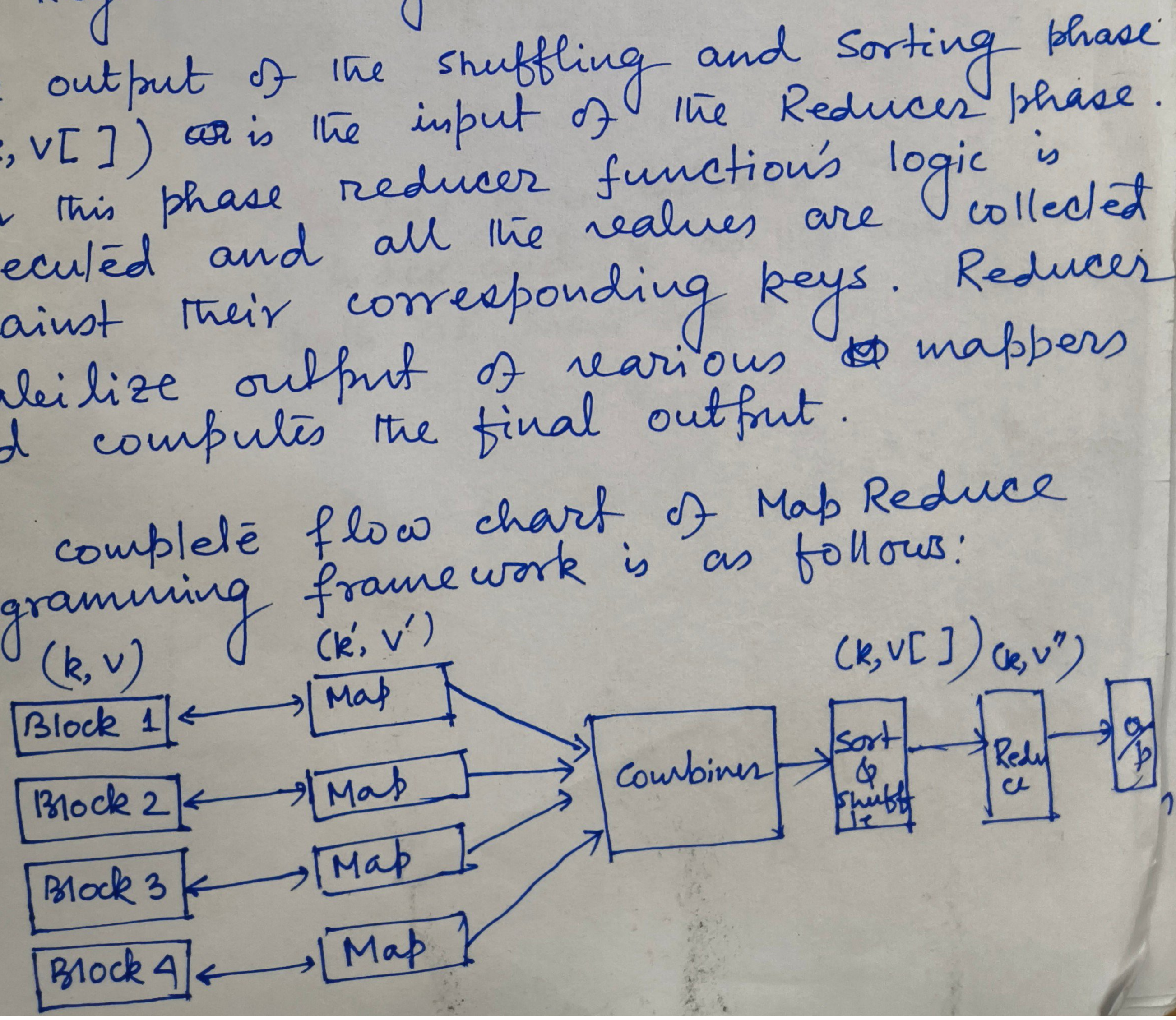
Reduce Phase

The second part of ~~map~~ MapReduce programming is the reduce phase. In the reduce phase the program takes the 'Output' from the Map as an input and reduces that data set into a smaller and more meaningful data set.

In between the 'Map' and the 'Reduce' phase there exist the shuffling and sorting phase. The output of various mapping parts (k', v') then goes into shuffling and sorting phase. All the same values are ~~delea~~ deleted and different values are grouped together based on same keys. The output of the shuffling and sorting phase ~~to~~ generates a key-value pair in the form of key and array of values (k, V[ ]).

The output of the shuffling and sorting phase (k, V[ ]) ~~ar~~ is the input of the Reducer phase. In this phase reducer function's logic is executed and all the values are collected against their corresponding keys. Reducer stabilize output of various mappers and computes the final output.

The complete flow chart of Map Reduce programming framework is as follows:



Matrix Multiplication using MapReduce

Let us take the example of Matrix Multiplication algorithm using MapReduce.

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ & $B = \begin{bmatrix} 2 & 1 \\ 5 & 1 \end{bmatrix}$ are two

2 × 2 matrix

The first matrix A has four elements and they are indexed as—

i as the row index and j as the column index. i and j both takes the value 0, 1.

$$\begin{array}{c|c} i \backslash j & \begin{matrix} 0 & \quad 1 \end{matrix} \\ \hline \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \end{array}$$

$\therefore$ $A_{00} = 1$ $A_{01} = 2$ $A_{10} = 3$ and $A_{11} = 4$.

Similarly the matrix B is indexed as j as the ~~row index~~ row index and k as the column index and both of them takes the value 0 and 1.

$B_{00} = 2$; $B_{01} = 1$; $B_{10} = 5$; $B_{11} = 1$.

$$\begin{array}{c|c} & k \\ \hline j & \begin{bmatrix} 2 & 1 \\ 5 & 1 \end{bmatrix} \end{array}$$

~~Note~~

Note that the same variable j is used to represent the column index of the matrix A and the row index of matrix B. The reason behind this is to ensure the rule of matrix multiplication —

The no of column of the first matrix must be equal to the no of row in the second matrix.

Finally the index of the product matrix will be $(i \times k)$

Step-1 Map task

In this phase we need to convert the elements of matrices A and B into key-value pair form.

The rule for Matrix A is —

$$\underbrace{(i, k)}_{\text{key}} \ \underbrace{(A, j, A_{ij})}_{\text{value}} .$$

Here i is the row index. k is a variable which takes the value '0' and '1' for a 2×2 matrix. j is the column index and A_{ij} is the value of the matrix element.

The rule for the matrix B is

$$\underbrace{(i, k)}_{\text{key}} \ \underbrace{(B, j, B_{jk})}_{\text{value}}$$

here k is the column index. 'i' is a variable which takes the value 0 and 1 for 2×2 matrices. j is the row index.

For the first matrix A

The first ~~a~~ element is —

A_{00} $\therefore$ $((\overset{i}{0}, \overset{k}{0}), (\overset{j}{A}, \overset{A_{ij}}{0}, 1))$

$i=0, j=0$

and $((0, 1)\ (A, 0, 1))$

Similarly

$A_{01} = 2$ $((0, 0)\ (A, 1, 2))$

$i=0, j=1$ $((0, 1)\ (A, 1, 2))$

$A_{10} = 3$ $((1, 0), (A, 0, 3))$

$i=1, j=0$ $((1, 1), (A, 0, 3))$

$A_{11} = 4$ $((1, 0), (A, 1, 4))$

$i=1, j=1$ $((1, 1), (A, 1, 4))$

For Matrix B.

$B_{00} = 2$ $((0, 0), (B, 0, 2))$

$j=0, k=0$ $((1, 0), (B, 0, 2))$

$B_{01} = 1$ $((0, 1), (B, 0, 1))$

$j=0, k=1$ $((1, 1)\ (B, 0, 1))$

$B_{10} = 5$ $((0, 0), (B, 1, 5))$

$j=1, k=0$ $((1, 0)\ (B, 1, 5))$

$B_{11} = 1$ $((0, 1), (B, 1, 1))$

$j=1, k=1$ $((1, 1), (B, 1, 1))$

Combiner Task

In this phase we will combine all the element having same (key, value) index into one array

$[(0,0), (A,0,1); (A,1,2); (B,0,2); (B,1,5)]$

Similarly $[(0,1), (A,0,1); (A,1,2); (B,0,1); (B,1,1)]$

$[(1,0), (A,0,3); (A,1,4); (B,0,2); (B,1,5)]$

$[(1,1), (A,0,3); (A,1,4); (B,0,1); (B,1,1)]$

(3) Reduced Task

For (0,0) check the j values inside all the members which are for this case $(A,\underline{0},1)$ $(A,\underline{1},2)$; $(B,\underline{0},2)$; $(B,\underline{1},5)$.

whenever we find the j values are same we write them one below the other for example in the present case $\{(A,\underline{0},1), (B,\underline{0},2)\}$ and $\{(A,1,2),(B,1,5)\}$ having the same j value 0 and 1 respectively.

$$\therefore \text{ For } (0,0) \longrightarrow \left\{\begin{matrix}(A, 0, \boxed{1}) \\ (B, 0, \boxed{2})\end{matrix}\right\} \left\{\begin{matrix}(A, 1, \boxed{2}) \\ (B, 1, \boxed{5})\end{matrix}\right\}$$

Now multiply the third parameters and add them together to get

$1 \times 2 = 2$ & $2 \times 5 = 10$

$\therefore F_{00} = 2 + 10 = 12$

This is the value of the (0,0) element of the product matrix

$$\therefore F_{00} = 12.$$

Similarly -

$[(0,1), (A,0,1); (A,1,2); (B,0,1); (B,1,1)]$

$$(0,1) \longrightarrow \left\{ \begin{matrix} \left(A, 0, \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) \\ (B, 0, \end{matrix} \right\} \left\{ \begin{matrix} \left(A, 1, \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right) \\ (B, 1, \end{matrix} \right\}$$

$$\therefore f_{0,1} = 1\times1 + 2\times1 = 1+2 = 3.$$

$[(1,0), (A,0,3); (A,1,4); (B,0,2); (B,1,5)]$

$$\therefore (1,0) \rightarrow \left\{ \begin{matrix} \left(A, 0, \begin{bmatrix} 3 \\ 2 \end{bmatrix}\right) \\ (B, 0, \end{matrix} \right\} \left\{ \begin{matrix} \left(A, 1, \begin{bmatrix} 4 \\ 5 \end{bmatrix}\right) \\ (B, 1, \end{matrix} \right\}$$

$$\therefore F_{10} = 3\times2 + 4\times5 = 26.$$

$[(1,1), (A,0,3); (A,1,4); (B,0,1); (B,1,1)]$

$$\therefore (1,1) \rightarrow \left\{ \begin{matrix} \left(A, 0, \begin{bmatrix} 3 \\ 1 \end{bmatrix}\right) \\ (B, 0, \end{matrix} \right\} \left\{ \begin{matrix} \left(A, 1, \begin{bmatrix} 4 \\ 1 \end{bmatrix}\right) \\ (B, 1, \end{matrix} \right\}$$

$$f_{11} = 3\times1 + 4\times1 = 3+4 = 7$$

1. The product matrix is –

$$\begin{bmatrix} f_{00} & f_{01} \\ f_{10} & f_{11} \end{bmatrix} = \begin{bmatrix} 12 & 3 \\ 26 & 7 \end{bmatrix}$$